

RESEARCH PROFILE

PhD researcher in Electrical Engineering at IIT Bombay working on Home Energy Management Systems (HEMS) for residential buildings. My research combines optimization, forecasting, data-driven methods, and physics-based modeling for battery-aware energy scheduling, demand response, and flexible cooling loads. I am interested in applying AI and data-driven methods to strengthen forecasting and control in flexible, grid-interactive residential energy systems.

RESEARCH INTERESTS

- Home Energy Management Systems (HEMS)
- Energy Forecasting
- AI and Data-Driven Energy Systems
- Optimization and Scheduling
- Battery Energy Storage
- Battery Degradation Modeling
- Demand Response and Demand Flexibility
- Smart Buildings and HVAC Control
- Smart Meter Data Analytics
- Renewable Energy Integration

EDUCATION

Dec 2023 – Present	PhD in Electrical Engineering <i>Indian Institute of Technology Bombay (IIT Bombay)</i> Focus: Battery-aware HEMS Optimization, Energy Scheduling, Demand Response Flexibility. Advisor: Prof. Anupama Kowli
2021 – 2023	M.Tech in Power Systems <i>National Institute of Technology, Kurukshetra</i> Thesis: Short-Term Forecasting for Power System Operation. CGPA: 8.83
2015 – 2019	B.Tech in Electronics and Electrical Engineering <i>Kalinga Institute of Industrial Technology, Bhubaneswar</i> CGPA: 7.29

RESEARCH EXPERIENCE

PhD	Battery-aware HEMS scheduling <i>IIT Bombay</i> - MILP-based HEMS scheduling integrating PV, battery, EV, and grid interaction, with multiple battery degradation formulations (quadratic, DoD-based linear, net-load-factor-based dynamic). - Multiobjective optimization balancing energy cost and battery degradation using weighted-sum and AUGMECON epsilon-constraint methods, including Pareto front and knee-point analysis.
PhD	Residential AC demand flexibility <i>IIT Bombay</i> - Experimental and simulation-based assessment of residential AC demand flexibility using inverter and non-inverter systems under Mumbai's tropical humid climate. - Developing a hybrid data-driven and physics-based inverter AC model, combining data-driven parameter identification with an equivalent thermal parameter (ETP) framework.

Aug – Dec 2023

Real-time HEMS simulator

Project Research Associate, IIT Bombay · with Enphase

- Developed a Python-based real-time simulator for Home Energy Management Systems.
- Integrated distributed energy resources including PV, battery storage, SG-ready heat pumps, electric vehicles, residential loads, and grid interaction.
- Modeled Li-ion battery operation under constant-current/constant-voltage (CCCV) and constant-power charging and discharging modes.

2021 – 2023

Load and price forecasting

M.Tech, NIT Kurukshetra

- Developed LSTM and GRU recurrent neural network models for short-term electricity-load forecasting and electricity market clearing price prediction.
- Investigated the effect of training optimizers, including Adam, Nadam, SGD, and RMSprop, on forecasting performance.

■ PUBLICATIONS

2025

[1] **Dey, M.K.**, Kowli, A. (2025). “Impact of Battery Degradation Modeling on Residential Energy Scheduling Performance”. In: 2025 IEEE International Conference on Energy Technologies for Future Grids (ETFG), IEEE, pp. 1–6. [IEEE Xplore ↗](#)

2024

[2] Somawanshi, A.D., **Dey, M.K.**, Kowli, A., Gugaliya, J., Choi, B., Nelakuditi, N.S. (2024). “Integration of Smart Grid Ready Heat Pumps in Home Energy Management Systems”. In: 2024 IEEE PES Innovative Smart Grid Technologies Asia (ISGT Asia), IEEE, pp. 1–6. [IEEE Xplore ↗](#)

2024

[3] **Dey, M.K.**, Chanana, S. (2024). “Comparison of LSTM and GRU for Predicting Market Clearing Price in Open Electricity Market”. In: Kumar, A., Singh, S.N., Kumar, P. (eds), Decarbonisation and Digitization of the Energy System, SGESC 2023, Lecture Notes in Electrical Engineering, vol. 1099, Springer, Singapore. DOI: 10.1007/978-981-99-7630-0_23. [Springer ↗](#)

2022

[4] **Dey, M.K.**, Chanana, S. (2022). “Short-term forecasting of a univariate time series load using LSTM based RNN model”. In: 2022 International Conference on Futuristic Technologies (INCOFT), IEEE, pp. 1–6. [IEEE Xplore ↗](#)

■ PRESENTATIONS & POSTERS

Poster

Experimental Assessment of Residential Cooling Flexibility for Demand Response in Mumbai

Rising Talents Award

ACE Conference

Dey, M.K., Kalojiya, G., Sankaran, S., Kowli, A.

Poster · 2026

Smart Meter Data Analytics: Decoding Electricity Meter Readings

India AI Impact Summit 2026

Presented the group’s work on smart-meter data analytics, including non-intrusive load monitoring (NILM), consumer profiling, and privacy-preserving synthetic data generation.

■ TEACHING EXPERIENCE

Current

Teaching Assistant, Energy Data Analytics

IIT Bombay

Past

Teaching Assistant, Machines Laboratory
IIT Bombay

Past

Teaching Assistant, Power Engineering II
IIT Bombay

■ TECHNICAL SKILLS

Python

MATLAB

C

C++

Pyomo

Gurobi

SCIP

Optimization

Machine Learning (LSTM/GRU)

Battery Modeling

HVAC Modeling

■ ADDITIONAL TRAINING

Python for Machine Learning and Data Science

Energy and Power System Optimization using GAMS